

Name: _____

Class: _____

Date: _____

Lab 14: Simulation Station

Python Lab · Section 3.16 Simulations



Goal

Build simulations, change their assumptions, and evaluate what they can and can't tell us.

Warm-up: Predict

```
import random
heads = 0
for i in range(10):
    if random.randint(1, 2) == 1:
        heads += 1
print(heads)
```

Will this always print 5? Run it a few times.

AP Connection

- Simulations are abstractions that use random values to model variability.
- Simplifications can introduce bias.

Build It

Step 1. Coin flips: run 10, 100, 10,000 flips. What happens to the percent of heads?

```
flips = int(input("Flips: "))
heads = 0
for i in range(flips):
    if random.randint(1, 2) == 1:
        heads += 1
print(heads / flips * 100, "%")
```

Step 2. Lunch line: each minute, 0–3 students arrive and 2 are served. Simulate 60 minutes.

```
line = 0
for minute in range(60):
    line += random.randint(0, 3)
    line = max(0, line - 2)
print("Students still waiting:", line)
```

Step 3. Change ONE assumption (e.g., 3 students served per minute). Run it 5 times. Record the results.

Step 4. List two real-world details your lunch-line simulation leaves out. How might that bias the results?

Date:

Build It (continued) · Level Up

Name one benefit and one limitation of your simulation.